

Sage Filter Test Document

1. Basic Arithmetic

Error executing Sage code: 'SageRunner' object has no attribute 'execute'

```
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 open>, 'quit': Use quit() or Ctrl-D (i.e. EOF) to exit, 'exit': Use exit() or
 Ctrl-D (i.e. EOF) to exit, 'copyright': Copyright (c) 2001-2022 Python Software
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Copyright (c) 1991-1995 Stichting Mathematisch Centrum, Amsterdam. All Rights Reserved., 'credits': Thanks to CWI, CNRI, BeOpen.com, Zope Corporation and a cast of thousands for supporting Python development. See www.python.org for more information., 'license': Type `license()` to see the full license text, 'help': Type `help()` for interactive help, or `help(object)` for help about object.}

a = 4

b = 15

c = 1024

Results: $-2 + 2 = 4 - 5 \times 3 = 15 - 2^{10} = 1024$

2. Mathematical Functions

Error executing Sage code: 'SageRunner' object has no attribute 'execute'

Function: $f(x) = \cos(x) + \sin(x)$

Derivative: $f'(x) = \cos(x) - \sin(x)$

Integral: $\int f(x) dx = -\cos(x) + \sin(x) + C$

3. Plotting

3.1 Basic Plot

[Error generating plot: name 'torus' is not defined Traceback (most recent call last): File "/home/dzack/notes/teaching/sage_filter.sage", line 212, in process_sage_plot 'disk': disk, 'sphere': sphere, 'torus': torus, 'parametric_plot3d': parametric_plot3d, ^^^^^ NameError: name 'torus' is not defined]

3.2 3D Plot

[Error generating plot: name 'torus' is not defined Traceback (most recent call last): File "/home/dzack/notes/teaching/sage_filter.sage", line 212, in process_sage_plot 'disk': disk, 'sphere': sphere, 'torus': torus, 'parametric_plot3d': parametric_plot3d, ^^^^^ NameError: name 'torus' is not defined]

4. Linear Algebra

Error executing Sage code: 'SageRunner' object has no attribute 'execute'

Matrix multiplication: $A \times B = \begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix}$

Inverse of A: $A^{-1} = \begin{pmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{pmatrix}$

5. Number Theory

Error executing Sage code: 'SageRunner' object has no attribute 'execute'

is_prime_n = False

- Is 123456789 prime? False
- Prime factorization: $123456789 = 3^2 \cdot 3607 \cdot 3803$

6. Silent Computation

The 10th Fibonacci number is 55.

7. Error Handling

7.1 Syntax Error

Error executing Sage code: 'SageRunner' object has no attribute 'execute'

[Syntax Error: invalid syntax (<string>, line 2)]

7.2 Runtime Error

Error executing Sage code: 'SageRunner' object has no attribute 'execute'

[Error executing code: division by zero]

8. LaTeX Output

Error executing Sage code: 'SageRunner' object has no attribute 'execute'

latex_A = \left(\begin{array}{rr} 1 & 2 \\ 3 & 4 \end{array}\right)

LaTeX output: $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$

9. Conclusion

This document tests various Sage features including: - Basic arithmetic and variables - Mathematical functions and calculus - 2D and 3D plotting - Linear algebra - Number theory - Silent computation - Error handling - LaTeX output generation